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Quality of Life After Percutaneous Coronary Intervention or Medical Therapy for Chronic Total Coronary Occlusions

EUROCTO and DECISION-CTO Meta-Analysis

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ABSTRACT

BACKGROUND The benefit of percutaneous coronary intervention (PCI) for chronic total coronary occlusions (CTOs) to improve clinical symptoms and quality of life (QoL) as compared with optimal medical therapy (OMT) is still under debate because of the scarcity of available randomized trials (RCTs).

OBJECTIVES We evaluated the effect of PCI vs OMT in patients with a CTO and no concomitant coronary lesions in a post-hoc pooled analysis of 2 RCTs.

METHODS A total of 518 patients with a single CTO and no other significant coronary lesion were extracted from 2 RCTs, EUROCTO and DECISION-CTO, which had compared PCI vs OMT. Randomization to PCI or OMT was 1:1 in DECISION and 2:1 in EUROCTO. The clinical status was assessed by the Seattle Angina Questionnaire (SAQ) at baseline and after 12 months, and clinical events were monitored for 3 years.

RESULTS PCI was successful in 92.2%. On an intention-to-treat analysis, PCI appeared to be superior to OMT for the change of angina frequency scores between baseline and follow-up (12.2 vs 8.6; $P = 0.009$), QoL (19.5 vs 11.3; $P < 0.001$), and the SAQ summary score (13.8 vs 8.5; $P < 0.001$). For physical limitation, the difference was just at the level of the Bonferroni correction for multiple tests ($P = 0.01$). There was a wide variability of changes in SAQ scores. For QoL, the major determinant for a significant improvement was a low baseline score and the assignment to PCI, whereas gender, diabetes, or lesion complexity had no influence. During a mean follow-up of 3.1 years, the clinical endpoints of cardiac death or nonfatal myocardial infarction were similar in both groups (OMT vs PCI: 2.7% vs 5.1%; $P = 0.17$). The rates of stroke or hospitalization for bleeding were similar, and only target lesion revascularizations were more frequent with OMT (18.8% vs 10.6%; $P = 0.005$).

CONCLUSIONS In this post-hoc analysis from 2 RCTs of patients with a single CTO and no significant concomitant lesion, PCI achieved better improvement in QoL, angina frequency, and the SAQ summary score than OMT with no signal of excess harm regarding clinical endpoints. (EUROCTO [Evaluate the Utilization of Revascularization or Optimal Medical Therapy for the Treatment of Chronic Total Coronary Occlusions; NCT01760083] and DECISION-CTO [Drug-Eluting Stent Implantation Versus Optimal Medical Treatment in Patients With Chronic Total Occlusion; NCT01078051]) (JACC. 2026;■:■–■) © 2026 by the American College of Cardiology Foundation.

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ABBREVIATIONS AND ACRONYMS

CCS = chronic coronary syndrome

CTO = chronic total coronary occlusion

EQ-5D-3L = European Quality of Life-5 Dimensions

MI = myocardial infarction

OMT = optimal medical therapy

PCI = percutaneous coronary intervention

QoL = quality of life

RCT = randomized controlled trial

Chronic total coronary occlusions (CTOs) make up 15% to 20% of lesions in patients with a chronic coronary syndrome (CCS).[1,2] Despite the presence of collateral channels supplying the occluded territory, these are insufficient to fully prevent ischemia.[3,4] Moreover, through coronary steal, the donor artery territory may be affected as well.[5] Revascularization of the CTO would therefore not only restore antegrade flow to the CTO territory, but also reverse ischemia in the non-CTO donor territory.[6]

A negative prognostic impact of a CTO was observed in registries of acute myocardial infarction (MI) occurring in the non-CTO territory.[7-9] Also, with CCS, patients with a CTO appear to have a worse prognosis, as recently shown in a subgroup analysis from the ISCHEMIA (International Study of Comparative Health Effectiveness with Medical and Invasive Approaches) trial.[10] Even though numerous registries indicated a prognostic benefit of percutaneous intervention (PCI) as compared with optimal medical therapy (OMT),[11-14] this was not proven in a randomized controlled trial (RCT).[15]

Irrespective of the presence of a CTO, large RCTs in CCS have failed to show a prognostic benefit of PCI as compared with OMT,[16,17] but there is a clear benefit for clinical symptom relief.[18,19] Also, patients with a CTO seem to improve symptomatically after PCI,[20-22] but current guidelines do not recommend revascularization of a CTO for symptom relief.[23] This is based on conflicting results from the 2 largest RCTs: the EUROCTO (Evaluate the Utilization of Revascularization or Optimal Medical Therapy for the Treatment of Chronic Total Coronary Occlusions; NCT01760083) study showed improved symptom relief and quality of life after PCI as compared with OMT,[24] whereas the DECISION-CTO (Drug-Eluting Stent Implantation Versus Optimal Medical Treatment in Patients With Chronic Total Occlusion; NCT01078051) study showed no difference.[25] These discrepancies can be explained by the high rate of non-CTO intervention in DECISION-CTO after randomization, whereas in EUROCTO non-CTO PCI was performed before randomization.

To overcome the disparity of both studies regarding the impact of non-CTO PCI, the investigators of both trials merged patients in a post-hoc, hypothesis-generating pooled analysis to create the cleanest randomized-derived dataset currently available to address a focused question: What happens when the CTO is the only clinically relevant lesion, and patients are randomized to antianginal medication or PCI?

METHODS

The design, rationale, and results of the EUROCTO trial and the DECISION-CTO trial have been previously published.[24,25] Both trials were open-label, multicenter RCTs. In brief, the EUROCTO trial randomized in a 2:1 allocation PCI plus OMT and OMT alone in patients recruited in 28 European centers with expertise for CTO PCI. The DECISION-CTO trial was conducted at 19 sites in Korea, India, Indonesia, Thailand, and Taiwan with a 1:1 randomization for either PCI or no PCI for the qualifying CTO lesion plus OMT for both groups.

Both studies required evidence of ischemia or clinical symptoms with a CTO in a major coronary artery with a vessel diameter ≥ 2.5 mm (American Heart Association site map 1-3, 6, 7, and 11).[26] The study protocols were approved by the relevant ethical committees with written informed consent from all patients.

STUDY SUBSET. To eliminate any influence of additional coronary disease other than the CTO, only patients with a single CTO and no other significant lesion were extracted from both trials. To harmonize the entry criteria of both studies, in-stent CTOs were excluded as well as patients with an ejection fraction <30%. These criteria were fulfilled by 272 patients from EUROCTO and 246 patients from DECISION-CTO. A comparison of the original study endpoints and the modified endpoints for this post-hoc patient-level meta-analysis is summarized in Supplemental Table 1.

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Lesion complexity was assessed with the J-CTO score.[27] PCI was carried out with a bilateral approach visualizing the distal territory of the CTO through collateral filling. Antegrade and retrograde techniques were used as necessary. In case of procedural failure, additional interventional attempts were encouraged, including referral to bypass surgery. Angiographic success was defined as a final angiographic residual stenosis of <20% by visual estimate and TIMI flow grade 3 after implantation of a drug-eluting stent. Dual-antiplatelet therapy of aspirin and $P2Y_{12}$ inhibitors was advised for 12 months after PCI. Patients in both treatment groups received standard of care medication for lipid control, hypertension, and diabetes as well as medication in case of left ventricular dysfunction. In the OMT group, antianginal medication was adapted during follow-up, and efforts were made to avoid crossover to revascularization of the CTO.

STUDY ENDPOINTS. The primary efficacy endpoint of the EUROCTO trial was a change of health status between baseline and 12 months (see Health Status Assessment) with PCI vs OMT, with the 3-year incidence of MI or cardiac death as the primary safety endpoint. DECISION-CTO was originally designed to address the impact of PCI vs OMT on a primary composite endpoint of death from any cause, MI, stroke, or any revascularization (Supplemental Table 1). Additional health status assessments had been performed at baseline and at 1, 6, 12, 24, and 36 months.

For the present pooled analysis, the primary efficacy endpoint was change of health status at 12 months. The key safety endpoints were the incidence of cardiac death and MI at 3 years after randomization.

HEALTH STATUS ASSESSMENT. Health status was assessed by the Seattle Angina Questionnaire (SAQ), a 19-item questionnaire that measures 5 domains related to coronary artery disease: angina frequency, physical limitations, quality of life (QoL), angina stability, and treatment satisfaction.[28,29] Scores range from 0 to 100, with higher scores indicating fewer symptoms and better health status. The SAQ summary score as a way to combine the most relevant domains for CCS—physical limitation, angina frequency, and QoL—in 1 score was calculated from the available patient-level data sets.[30] Significant improvements in each domain are predefined for physical limitation (≥8 points), angina frequency (≥20 points), and QoL (≥16 points).[28,31] Because angina stability would require monthly assessment, which was not available in the EUROCTO trial, it is not included in this combined analysis.

The European Quality of Life-5 Dimensions (EQ-5D-3L) instrument[32] was assessed in both studies. It is a measure of mobility, self-care, usual activity, pain or discomfort, and anxiety or depression. The visual analog scale represents a linear scale from 0 to 100 representing the patient's general health status. The questionnaires were administered in each patient's native language and centrally evaluated by the clinical trial organizations.

ADVERSE EVENT DEFINITION AND ASSESSMENT. A major adverse cardiovascular or cerebrovascular event was defined as cardiac death, nonfatal MI, stroke, or ischemia-driven revascularization during follow-up. Criteria for acute MI followed the third universal definition of MI[33] and required a rise of cardiac biomarker values (cardiac troponin for the EUROCTO trial with ≥1 value above the 99th percentile upper reference limit or creatine kinase-myocardial band for the DECISION-CTO trial with elevations of >5 times the upper normal limit combined with either symptoms of ischemia, new significant ST-segment changes, new left bundle branch block, development of pathologic Q waves in the electrocardiogram, new regional wall motion abnormality, or identification of an intracoronary thrombus by angiography or autopsy). Stent thrombosis was defined according to the Academic Research Consortium criteria.[34]

STATISTICAL ANALYSIS. Sample size calculations for both studies and their initial patient cohort have been published.[24,25] Although both trials were terminated early because of slow recruitment, the statistical power of the sample size for the health status assessment was sufficient at a level of 90% to show superiority for either treatment arm.

The analysis is based on the intention-to-treat population. An as-treated calculation of the clinical endpoints was also performed, as was an analysis according to the success of revascularization. Mean ± SD are reported for continuous variables and compared with a Wilcoxon rank-sum test and numbers and percentages for categorical variables and compared by a chi-square test. The analysis of the primary endpoint was performed as follows. Each component of the SAQ was analyzed by means of an analysis of covariance with the differential score at 12 months as a response and baseline scores, study origin, J-CTO score, gender, presence of diabetes, and age as covariates. A Bonferroni correction for multiple testing of the domains and the summary score was applied. Hence, a 2-sided significance level

TABLE 1 Patient and Lesions Characteristics of Patients With a CTO Randomized to OMT or PCI

	All (n = 518)	OMT (n = 224)	PCI (n = 294)	*P* Value
Age, y	62.8 ± 10.1	62.3 ± 9.9	63.2 ± 10.2	0.37
Male	82.8 (429)	84.8 (190)	81.3 (239)	0.29
Body mass index, kg/m^2	27.0 ± 4.2	26.7 ± 3.5	27.3 ± 4.7	0.33
Diabetes	29.2 (151)	29.5 (66)	28.9 (85)	0.87
Hypercholesterolemia	80.5 (417)	80.4 (180)	80.6 (237)	0.94
Hypertension	68.5 (355)	65.2 (146)	71.1 (209)	0.15
History of smoking	68.1 (353)	66.1 (148)	69.4 (204)	0.47
Previous myocardial infarction	13.5 (70)	11.6 (26)	15.0 (44)	0.27
Previous bypass surgery	5.6 (29)	3.1 (7)	7.5 (22)	0.03
Previous PCI	34.8 (180)	33.0 (74)	36.1 (106)	0.47
Left ventricular ejection fraction, %	56.8 ± 9.8	56.9 ± 9.4	56.7 ± 10.0	0.84
No. of vessel disease				0.10
1	71.6 (371)	70.1 (157)	72.8 (214)	
2	18.7 (97)	22.3 (50)	16.0 (47)	
3	9.7 (50)	7.6 (17)	11.2 (33)	
Reversible ischemia tested	61.0 (316)	56.7 (127)	64.3 (189)	0.08
Viability tested	85.3 (442)	80.4 (180)	89.1 (262)	0.005
CTO location				0.73
Right coronary artery	51.5 (267)	50.5 (113)	52.4 (154)	
Left anterior descending artery	40.1 (208)	40.2 (90)	40.1 (118)	
Left circumflex artery	8.3 (43)	9.4 (21)	7.5 (22)	
J-CTO score	1.81 ± 0.75	1.73 ± 0.74	1.87 ± 0.78	0.05

Values are mean ± SD or % (n).
OMT = optimal medical therapy; PCI = percutaneous coronary intervention.

of 0.01 was used. For all other endpoints, a 2-sided significance level of 0.05 without correction for multiple testing was applied. The incidence of a significant improvement for an individual patient in physical limitation (≥8 points), angina frequency (≥20 points), and QoL (≥16 points) was compared by a chi-square test between both treatment groups.

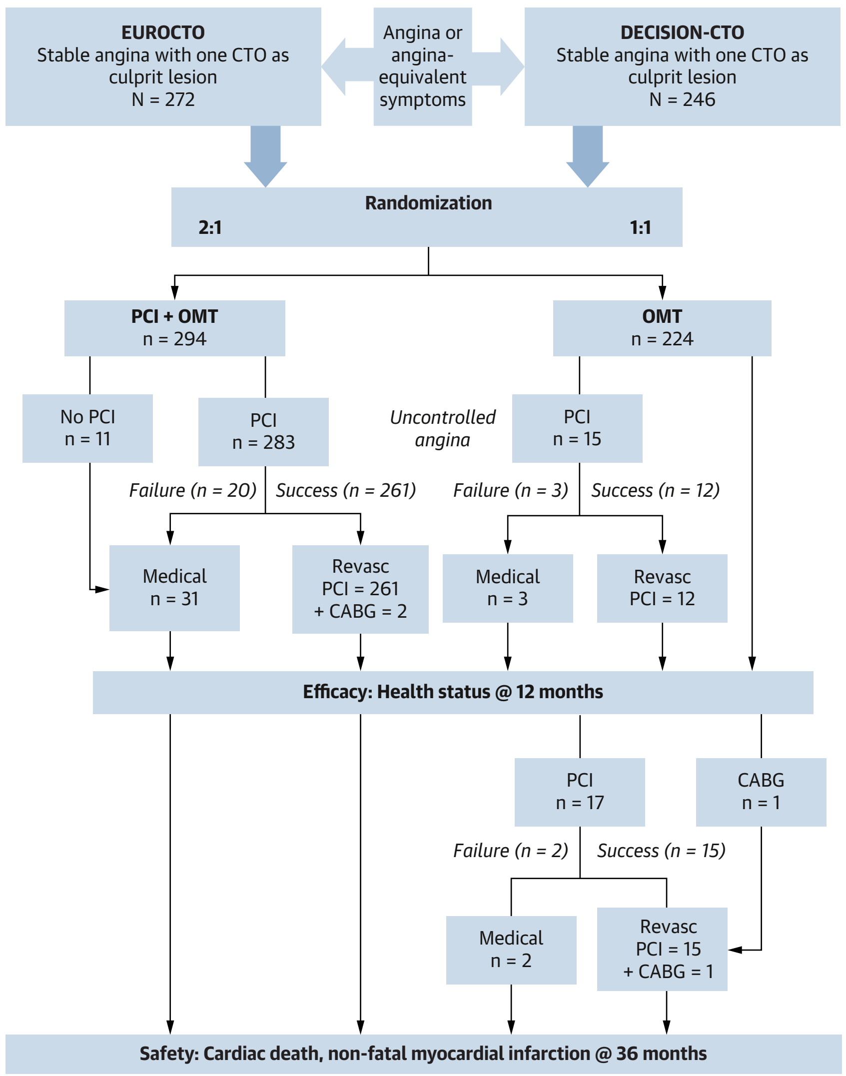
The EQ-5D-3L visual health state was analyzed as a secondary endpoint. Changes in EQ-5D-3L domains were compared by a proportional odds model, and the visual analog scale was assessed by analysis of covariance like the SAQ domains. The number of adverse events were compared between groups by means of a chi-square test of the Kaplan-Meier estimates. All analyses were performed with NCSS 2025 Statistical Software (NCSS, LLC), and MedCalc Statistical Software version 23.4.8 (MedCalc Software Ltd).

RESULTS

PATIENT RECRUITMENT AND CLINICAL CHARACTERISTICS. Of the 518 patients with a single CTO, 272 were included from EUROCTO and 246 from DECISION-CTO. European patients were older with a higher body mass index, more comorbidities and risk factors, and prior history of coronary events (Supplemental Table 2). The distribution of target vessels was different, with more right coronary artery occlusions in EUROCTO and more left anterior descending coronary occlusions in DECISION-CTO. The J-CTO score was higher in the EUROCTO patient group. Because of a 2:1 randomization ratio in EUROCTO and 1:1 in DECISION-CTO, 294 patients were in the PCI arm and 224 in the OMT arm (**Table 1**). Clinical characteristics did not show significant differences except for less prior bypass surgery and less viability tests performed in OMT patients. The J-CTO score was lower in the OMT group.

INTERVENTIONAL AND MEDICAL MANAGEMENT. Eleven patients randomized to PCI (3.7%) did not undergo PCI because of patient or investigator preference. In the remaining 283 patients, the PCI success rate was 88.7% for the first attempt and 92.2% after a second attempt in 15 patients, whereas 2 patients were referred to coronary artery bypass grafting (**Figure 1**). One-third of procedures required a retrograde technique (Supplemental Table 3). In the OMT group, 15 patients (6.7%) crossed over to PCI within the first year. Another 18 patients (8.0%) underwent CTO revascularization after 1 year with one of them

FIGURE 1 Diagram Showing Patient Recruitment From Original Studies and Assigned Initial Treatment Strategy and Coruse of Follow-Up



Study flow chart of the patient subset from the EUROCTO and DECISION-CTO trials. CABG = coronary artery bypass grafting; DECISION-CTO = Drug-Eluting Stent Implantation Versus Optimal Medical Treatment in Patients With Chronic Total Occlusion; EQ-5D = European Quality of Life-5 Dimensions; EUROCTO = Evaluate the Utilization of Revascularization or Optimal Medical Therapy for the Treatment of Chronic Total Coronary Occlusions; OMT = optimal medical therapy; PCI = percutaneous coronary intervention.

directly referred to coronary artery bypass grafting. PCI was successful in 84.3% of crossover patients (Supplemental Table 3). Eight patients in the PCI group and 3 in the OMT group withdrew their consent for further follow-up.

The components of concomitant medical therapy are summarized in Supplemental Table 4. At baseline, 53.3% of patients in the OMT group were receiving adenosine diphosphate antagonists, whereas it was 93.5% in the PCI group. Calcium antagonists and nitrates were prescribed more often in the OMT group, and this difference remained at follow-up. The number of antianginal medications was significantly higher with OMT as compared with PCI both at baseline and at follow-up.

CLINICAL AND HEALTH STATUS CHANGES IN THE STUDY GROUPS. Three SAQ domains showed higher scores at baseline in the OMT group as compared with the PCI group (Table 2). At follow-up, in an intention-

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TABLE 2 Seattle Angina Questionnaire Health Status Changes Between Baseline and 12 Months of Follow-Up in Patients With a CTO Randomized to OMT or PCI

	OMT (n = 224)	PCI (n = 294)	*P* Value	Cohen's d
Baseline assessments done	213 (95.1)	282 (95.9)		
Follow-up assessments done	188 (83.9)	238 (81.0)		
Physical limitation				
Baseline	80.1 ± 22.4 (77.1-83.2)	75.4 ± 23.0 (72.6-78.1)	0.02	
Follow-up	85.1 ± 20.0 (82.5-88.2)	85.1 ± 20.9 (82.4-87.9)	0.98	
Δ change[a]	5.6 ± 17.2 (3.1-8.1)	10.1 ± 17.1 (7.8-12.4)	0.01	0.26
Angina frequency				
Baseline	84.2 ± 20.8 (81.3-87.0)	79.1 ± 22.0 (76.6-81.7)	0.001	
Follow-up	90.9 ± 16.4 (88.5-93.2)	93.1 ± 14.6 (91.2-94.9)	0.15	
Δ change[a]	8.2 ± 14.3 (6.2-10.3)	12.5 ± 14.3 (10.7-14.4)	0.003	0.30
Treatment satisfaction				
Baseline	82.9 ± 14.7 (80.9-84.9)	82.6 ± 16.3 (80.7-84.5)	0.99	
Follow-up	84.9 ± 15.0 (83.0-87.2)	88.7 ± 14.2 (86.9-90.5)	0.06	
Δ change[a]	2.1 ± 13.8 (0.1-4.2)	5.2 ± 13.8 (3.4-7.0)	0.027	0.22
Quality of life				
Baseline	61.2 ± 23.8 (58.0-64.5)	55.2 ± 22.3 (52.5-57.8)	0.003	
Follow-up	70.9 ± 20.5 (68.0-73.9)	76.6 ± 21.9 (73.8-79.4)	0.006	
Δ change[a]	11.3 ± 19.8 (8.4-14.2)	19.5 ± 19.7 (17.0-22.2)	<0.001	0.42
Sum score				
Baseline	75.3 ± 18.7 (72.7-77.8)	69.8 ± 17.9 (67.7-71.9)	0.001	
Follow-up	82.0 ± 15.5 (79.8-84.2)	85.1 ± 14.8 (83.2-87.0)	0.04	
Δ change[a]	8.1 ± 13.6 (6.1-10.0)	14.2 ± 13.6 (13.2-15.9)	<0.001	0.45

Values are n (%) or mean ± SD (95% CI). *P* value for the difference between treatment groups as assessed by an analysis of covariance for each Seattle Angina Questionnaire item. The Δ change is considered significant if $P < 0.01$ to account for the Bonferroni correction for multiple tests. [a]Value of Δ change is corrected for the difference in baseline scores.

Abbreviations as in Table 1.

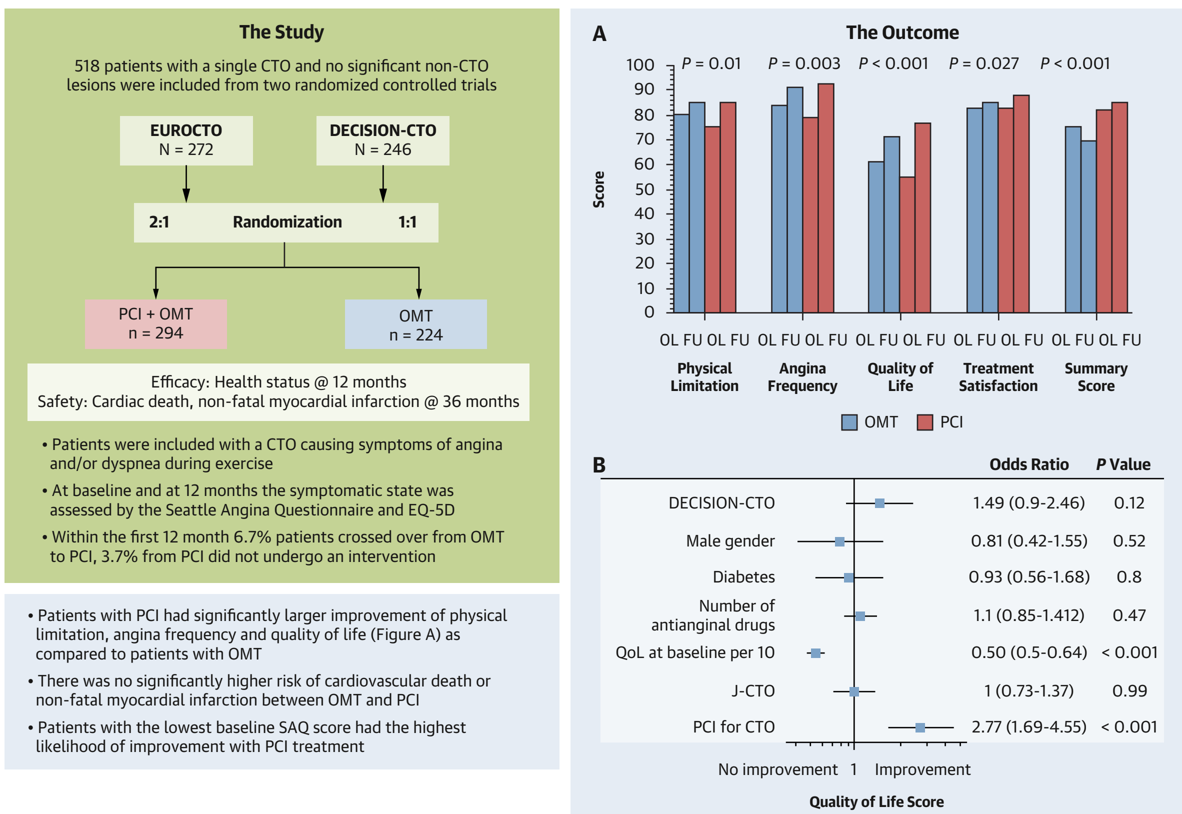
to-treat analysis, angina frequency and QoL improved significantly more in the PCI group than in the OMT group, with $P < 0.01$, the statistical level prespecified for the primary endpoint (Central Illustration). Also, the SAQ summary score improved significantly more in the PCI group as compared with the OMT group. The changes in the physical limitation score were just at the border of the Bonferroni correction with $P = 0.01$, whereas treatment satisfaction was below with a $P = 0.027$ (Table 2). The result was not influenced by J-CTO score, gender, age, or diabetes.

There was a slightly lower score at baseline in the OMT group for the EQ-5D-3L categories and a higher visual analog scale. The latter improved in the PCI group as compared with the OMT group, whereas the other categories showed no significant difference between the treatment arms (Table 3).

PREDICTING IMPROVEMENT IN HEALTH STATUS. The improvements of SAQ domains considered significant (physical limitation ≥8 points, angina frequency ≥20 points, QoL ≥16 points) were observed more often in the PCI group than in the OMT group for angina frequency (40.7% vs 26.5%, $P = 0.002$) and QoL (66.1% vs 41.9%, $P < 0.001$) (Figure 2). Although the overall score change for physical limitation was not significant at $P = 0.01$, there were still more significant individual improvements of physical limitation with PCI (52.6% vs 37.2%, $P = 0.004$). The higher incidence of improvements for QoL after PCI was consistent in both study subsets with a significant OR of 2.73 (95% CI: 1.59-4.71) in EUROCTO and 3.18 (95% CI: 1.69-5.99) in DECISION-CTO. For physical limitation, the OR was not significant for EUROCTO (1.65; 95% CI: 0.95-2.87) but was for DECISION-CTO (1.94; 95% CI: 1.06-3.56). The OR for angina frequency was significant for EUROCTO (2.25; 95% CI: 1.24-4.06) but not for DECISION-CTO (1.71; 95% CI: 0.92-3.18) (Supplemental Figure 1).

Figure 3 depicts the wide variability of these significant score changes. For physical limitation, no significant change was observed in 44.4% with OMT and 34.4% with PCI (Figure 3A). For angina frequency, no significant change was observed in most

CENTRAL ILLUSTRATION Quality of Life After Percutaneous Coronary Intervention or Medical Therapy for Chronic Total Occlusions



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The subset meta-analysis of 2 randomized trials of patients with a single CTO and non-CTO. Changes in the SAQ domains from baseline to follow-up with OMT or PCI. Higher scores indicate less symptomatic impairment (A). Forrest plot for the possible determinants of a significant improvement of the QoL domain (≥16 points) from baseline to 12 months follow-up. The baseline score and the assigned treatment arm were the only significant determinants (B). CTO = chronic total coronary occlusion; DECISION-CTO = Drug-Eluting Stent Implantation Versus Optimal Medical Treatment in Patients With Chronic Total Occlusion; EQ-5D = European Quality of Life-5 Dimensions; EUROCTO = Evaluate the Utilization of Revascularization or Optimal Medical Therapy for the Treatment of Chronic Total Coronary Occlusions; OMT = optimal medical therapy; PCI = percutaneous coronary intervention; QoL = quality of life.

patients, 63.8% with OMT and 53.4% with PCI. Worsening angina was infrequent (9.7% vs 5.9%) (Figure 3B).

The highest number of improvements were observed for QoL in the PCI arm (Figure 3C). Worsening QoL scores were more frequent with OMT than PCI (17.4% vs 6.9%). Patients with an improved QoL had a baseline score of 47 ± 19, whereas those without an improvement had a considerably higher baseline score of 72 ± 22 ($P < 0.001$). Figure 4 demonstrates the relation between the baseline QoL score and the extent of QoL score changes at follow-up. The slope of the regression lines for this correlation was comparable for OMT (-0.65) and PCI (-0.62). Although the number of patients with significant improvement was more frequent with PCI, there were also worsening scores after PCI. Aside from the baseline score, an improvement of QoL (≥16 points) was determined by assignment to the PCI arm but not by gender, presence of diabetes, J-CTO score, number of antianginal drugs, and study origin (Central Illustration).

CLINICAL SAFETY ENDPOINTS. In the PCI group, 3 non-Q-wave MIs were observed (1.0%) in 283 procedures, all with unsuccessful procedures. No patient required emergency surgery. Two patients had transient cerebral ischemic events (0.7%) after PCI. Two

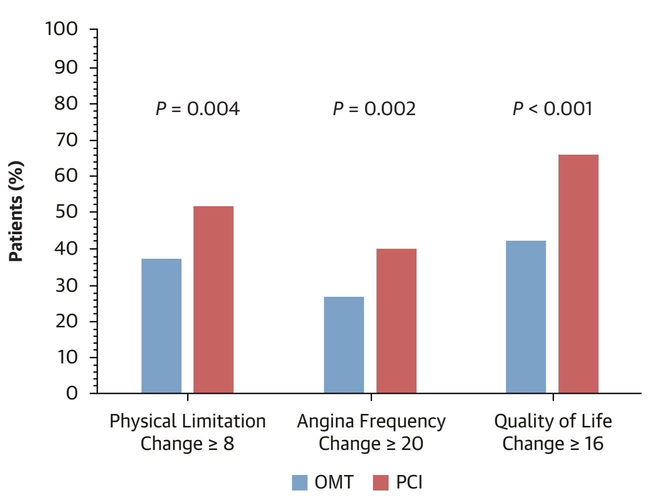
TABLE 3 EQ-5D-3L Health Status Changes at 12 Months in Patients With a CTO Randomized to OMT or PCI

	OMT (n = 224)	PCI (n = 294)	*P* Value
EQ-5D visual analog scale			
Baseline	69.9 ± 17.9 (215)	66.5 ± 18.6 (286)	0.04
Follow-up at 12 mo	75.6 ± 15.6 (201)	76.0 ± 16.6 (254)	0.78
Δ change[a]	6.4 ± 15.4	9.6 ± 15.4	0.03
EQ-5D mobility			
Baseline	71.2 / 27.8 / 0.5 (216)	66.4 / 32.9 / 0.7 (286)	0.45
Follow-up at 12 mo	76.8 / 23.2 / 0 (198)	80.7 / 19.1 / 0 (256)	0.29
Improved/unchanged/worsened	14.0 / 75.1 / 10.9	20.1 / 73.2 / 6.7	0.12
EQ-5D self-care			
Baseline	92.6 / 6.5 / 0.9 (215)	89.2 / 10.1 / 0.7 (286)	0.35
Follow-up at 12 mo	94.5 / 5.5 / 0 (199)	92.6 / 7.0 / 0.4 (256)	0.54
Improved/unchanged/worsened	5.7 / 90.7 / 3.6	5.1 / 91.3 / 3.5	0.61
EQ-5D activities			
Baseline	73.5 / 25.1 / 1.4 (215)	66.4 / 30.4 / 3.2 (286)	0.16
Follow-up at 12 mo	77.4 / 21.6 / 1.0 (199)	80.9 / 20.0 / 1.2 (256)	0.62
Improved/unchanged/worsened	11.9 / 79.4 / 8.8	23.6 / 68.5 / 7.9	0.08
EQ-5D pain/discomfort			
Baseline	49.8 / 47.9 / 2.3 (215)	47.4 / 50.2 / 2.5 (285)	0.87
Follow-up at 12 mo	68.7 / 30.3 / 1.0 (198)	70.7 / 28.1 / 1.2 (256)	0.87
Improved/unchanged/worsened	26.4 / 65.8 / 7.8	33.2 / 57.3 / 9.5	0.32
EQ-5D anxiety/depression			
Baseline	66.5 / 32.1 / 1.4 (216)	66.4 / 28.7 / 4.9 (286)	0.09
Follow-up at 12 mo	75.4 / 23.1 / 1.5 (199)	75.8 / 23.4 / 0.8 (256)	0.76
Improved/unchanged/worsened	20.6 / 67.0 / 12.4	24.0 / 64.6 / 11.4	0.20

Values are mean ± SD (n) or percentage improved / unchanged / worsened (n). Follow-up after correction for baseline for EQ-5D-3L for visual analog scale is mean ± SD assessed by analysis of covariance with the baseline value and the randomized treatment as independent covariates. [a]Value of Δ change is corrected for the difference in baseline scores. Percentages in each category at baseline and 12 months follow-up are reported for the subdomains. The *P* value determined by chi-square test.

EQ-5D-3L = European Quality of Life-5 Dimensions; other abbreviations as in Table 1.

FIGURE 2 Incidence of Significant SAQ Domain Changes With OMT or PCI



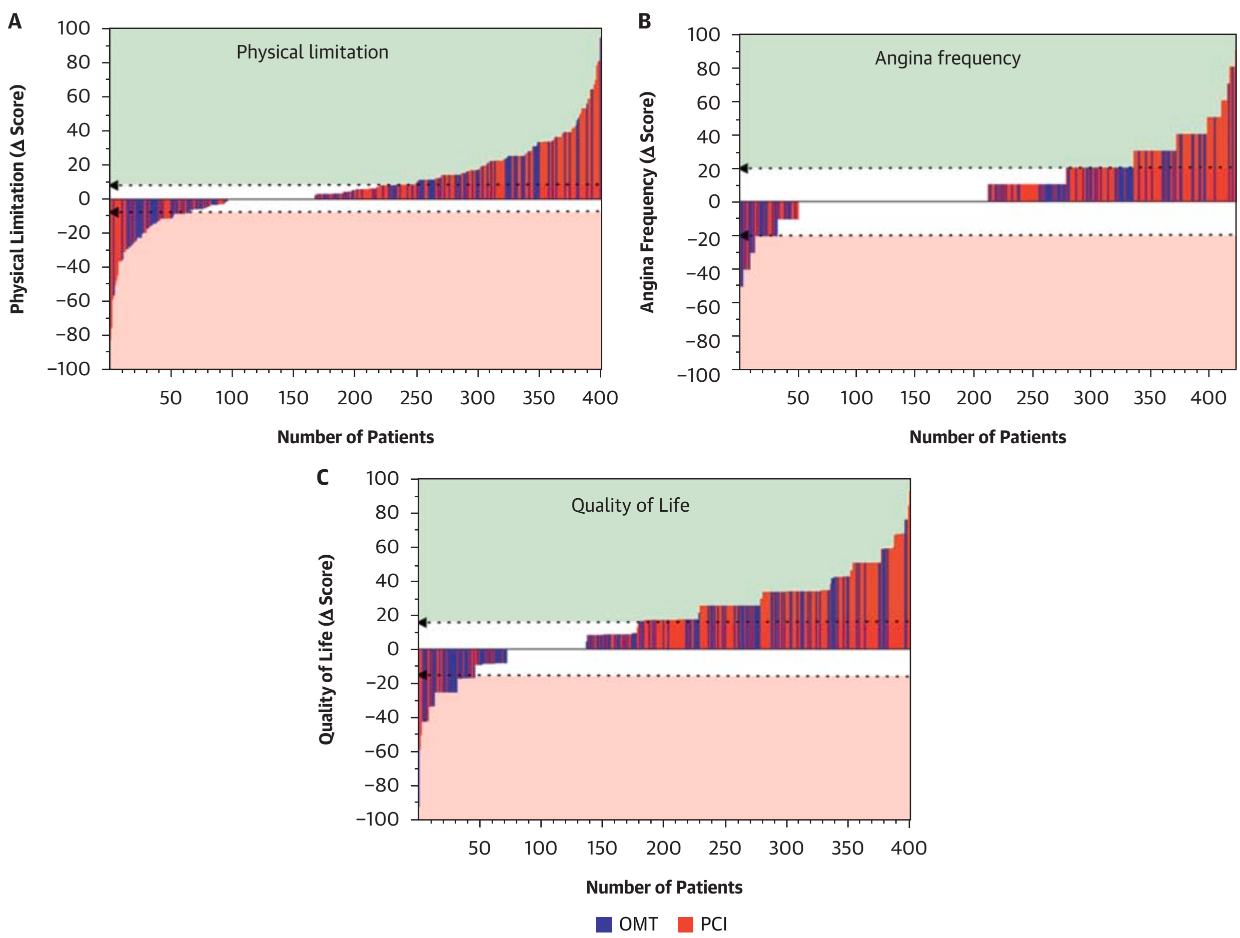
Comparison of clinically relevant changes in symptoms and quality of life with medical or interventional treatment of a single CTO. Abbreviations as in Figure 1.

access site complications required blood transfusion (0.7%), and 2 cases of tamponade after coronary perforation (0.7%) were successfully treated with pericardiocentesis. The median radiation dose was 2.45 Gy. No patients had postprocedure renal failure.

During a mean follow-up of 3.1 ± 0.6 years, the incidence of cardiovascular death or nonfatal MI was not significantly different in either group (2.7% in the OMT group and 5.1% in the PCI group; log-rank $P = 0.17$) (Table 4, Figure 5). Aside from 3 procedure-related MIs (1.0%), the spontaneous MI rate during follow-up was 2.0%, whereas it was 1.3% in the OMT group. During crossover procedures, no periprocedural MI was observed. An as-treated analysis is presented in Supplemental Figure 2.

The incidence of major cardiovascular and cerebrovascular events was higher in the OMT group (18.8%) than in the PCI group (10.6%; $P < 0.005$) because of more ischemia-driven revascularization in the OMT group (Table 4). In the PCI group, 8 patients with an initially successful procedure had a lesion recurrence requiring repeat PCI (2.7%), 2 of them

FIGURE 3 **The Spectrum of Individual Changes of SAQ Domains**



Waterfall plots showing the individual improvements of the Seattle Angina Questionnaire (SAQ) domains (A) physical limitation, (B) angina frequency, and (C) quality of life (QoL). Colored columns indicate the treatment arm, with the PCI arm predominantly observed on the right end of the spectrum. Shaded areas indicate the limits for significant improvements in the respective domains (≥8 for physical limitation, ≥20 for angina frequency, ≥16 for QoL).

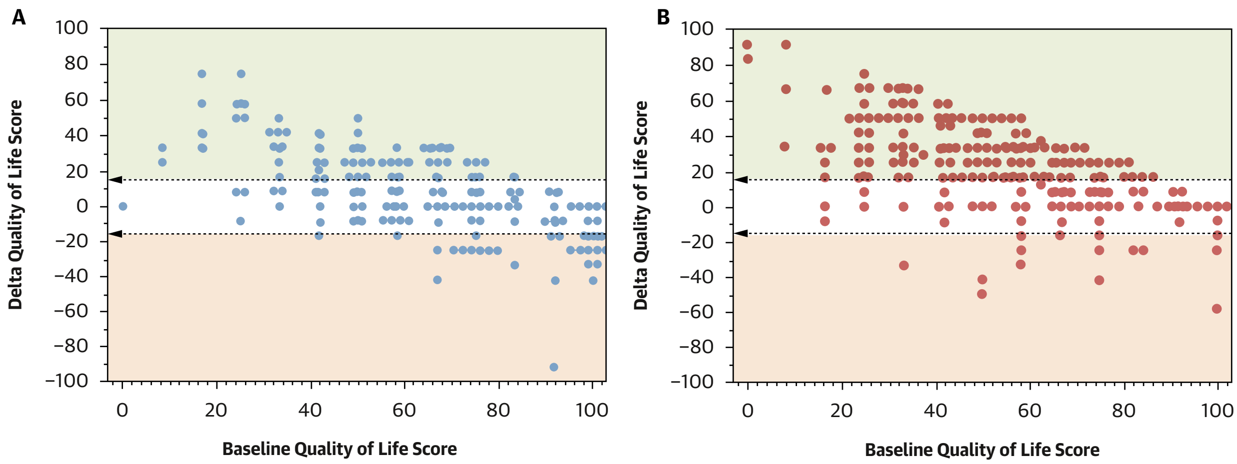
with late stent thrombosis (0.7%). One patient with an initially failed procedure underwent coronary artery bypass grafting. In addition, 6 revascularizations occurred in the PCI group (2.0%) because of disease progression in a previously untreated segment. Stroke events occurred in 5 patients in both groups, of which 2 were periprocedural in the PCI group.

DISCUSSION

In this combined, pooled patient-level meta-analysis from 2 RCTs, EUROCTO and DECISION-CTO, PCI appeared to be superior to OMT alone for symptom relief and improved QoL in patients with a single CTO and no significant non-CTO lesion. This was observed in 2 SAQ domains and the SAQ summary score. There was no significant difference in clinical safety endpoints of cardiac death or MI, but because of the low incidence of endpoints and limited patient numbers, we cannot draw any prognostic conclusions. The incidence of major cardiovascular and cerebrovascular events was higher with OMT, but this was largely driven by crossover from OMT to PCI and does not of itself necessarily indicate an important difference between the treatment arms.

SYMPTOMATIC IMPROVEMENT AND QoL AFTER CTO PCI. The combination of patients with predominantly single-vessel disease, normal left ventricular function, and low rate of prior coronary interventions resulted in higher baseline scores across the SAQ domains as compared with those observed in all-comer registries,[21,35] indicating an inclusion bias for less symptomatic patients well known for RCTs in

FIGURE 4 Dependence Between Extent of Change of QoL With Baseline Score



Relation between the baseline score and the change of scores at 12 months follow-up for quality of life in the (A) optimal medical therapy and (B) percutaneous coronary intervention treatment arm. The lower the baseline score, the higher the potential increase, with the shaded area depicting those patients with a significant improvement (≥16 points; green) or worsening (≤16 points; red). Individual points in the same category are stacked horizontally.

patients with a CTO.[15] Highly symptomatic patients will often refuse participation or are not put forward by their cardiologist for randomization to OMT when there is the option of a revascularization therapy.

Despite the inclusion of less symptomatic patients, those receiving PCI seemed to benefit regarding their QoL and clinical symptoms as compared with anti-anginal therapy alone. However, there was a wide variability of individual responses. To discriminate who would benefit, the consistent observation was that those with the lowest SAQ domain scores at baseline improved the most. This was consistent for QoL, angina frequency, and SAQ summary score and emphasizes the importance of the careful assessment of the symptomatic status of potential candidates for CTO PCI. With this wide variability of individual responses, PCI increased the probability of benefit in symptomatic patients but with real uncertainty at the individual level.

EVIDENCE FROM RCTs AND REGISTRIES FOR BETTER SYMPTOM CONTROL WITH REVASCULARIZATION IN CCS. It is an ongoing debate whether revascularization should be the primary option for patients with CCS. The current European and U.S. guidelines do not recommend it as a routine approach based on several

TABLE 4 Adverse Events at Follow-Up of 36 Months in Patients With a CTO Randomized to OMT or PCI (Intention-to-Treat Analysis)

Intention-to-Treat Analysis	OMT (n = 224)	PCI (n = 294)	*P* Value
Primary endpoint	2.7 (0.5-4.8) [6]	5.1 (2.6-7.6) [15]	0.17
Cardiac death	1.3 (0-2.9) [3]	2.0 (0.4-5.2) [6]	0.55
Nonfatal myocardial infarction	1.3 (0-2.9) [3]	3.4 (1.3-5.5) [9]	0.20
Periprocedural myocardial infarction	–	1.0 (0-2.2) [3]	
Noncardiac death	1.3 (0-2.9) [3]	2.0 (0.4-5.2) [6]	0.55
Major adverse cardiovascular and cerebrovascular events	18.8 (13.6-23.9) [38]	10.6 (7.0-14.1) [26]	0.005
Ischemia-driven revascularization	14.7 (10.0-19.4) [33]	5.4 (2.8-8.1) [16]	0.001
Stroke	2.2 (0.2-4.2) [5]	1.7 (0.2-3.2) [5]	0.66

Values are % (95% CI) [n].
Abbreviations as in Table 1.

RCTs.[23,36] However, a symptomatic benefit has been shown for many of these studies.[18,19] The recent subgroup analysis from the ISCHEMIA trial confirmed that observation also for patients with a CTO.[10] Although revascularization did not influence clinical endpoints, it improved the symptomatic status and QoL.[10] CTO PCI is still considered a special subset of interventional therapy because, historically, with lower success rates and higher complications rates.[37,38] However, these limitations are overcome with risk and success rates comparable with complex non-CTO lesions in expert centers,[39,40] as confirmed in the present study.

The potential benefit of PCI observed in this study in a less symptomatic RCT population could translate into a greater benefit in more symptomatic patients from registries.[21,41,42] However, this meta-analysis of EUROCTO and DECISION-CTO also did not provide any evidence of a prognostic benefit of PCI in this population of rather mildly symptomatic patients.

There is no doubt that PCI must be combined with OMT for lipid-lowering components as well as control of hypertension and diabetes, all with clear prognostic indication in the context of CCS.[36] Intensified antithrombotic treatment after stent placement in addition is advised for a year after complex PCI and did not result in major bleeding events in the current study. What is left out of the discussion on medical management as the "gold standard" and a supposedly safer substitute for interventional therapy is the fact that a medical approach to control symptoms increases the number of drugs to be taken by the patients for an indefinite time and that these additional nonprognostic drugs may likely increase the spectrum of side effects and increase problems with compliance.[43]

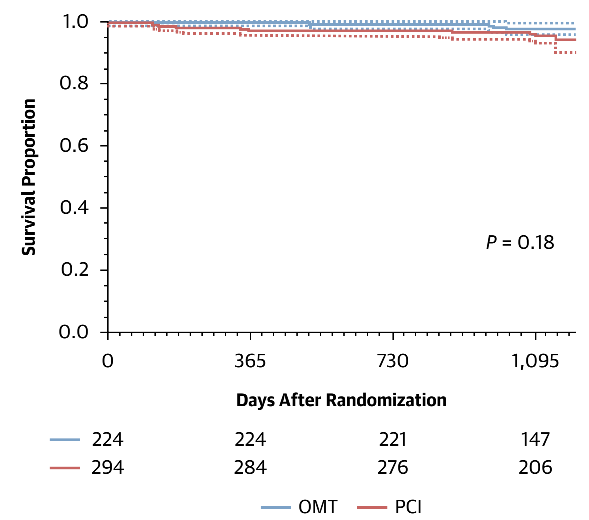
STUDY LIMITATIONS. This study consists of a subset of patients from 2 different RCTs with different ethnicity and risk factor prevalence. Coronary disease was less extensive in the younger Korean group, and patients with silent ischemia were also included, which may explain the higher baseline scores for health impairment indicating less physical limitation and less angina, but QoL was similarly impaired in both studies. Although the merger of patients does not create a new RCT, the patient selection in this meta-analysis allows the elimination of the confounding influence of non-CTO PCI that was present in the original studies either before (EUROCTO) or after (DECISION-CTO) randomization. In both studies, there was an imbalance in baseline SAQ scores with lower scores (ie, more symptomatic patients assigned to PCI). In addition, the follow-up SAQ assessment was missing in about one-fifth of patients. The EQ-5D-3L subcategories showed no significant difference between treatment groups except for the visual analog scale, indicating that this patient-reported outcome measure was less sensitive to treatment effects than SAQ in this patient group of low symptom levels.

Given the open-label design, expectancy effects likely influence all symptom endpoints so that the observed benefit represents a combination of physiological and expectancy-related effects that cannot be disentangled in these data. Also, the awareness of a successful treatment of the CTO may influence the subjective patient self-evaluation.

The limitation to expert centers led to high procedural success, but these results may not be reproduced in less experienced hands. Even though the patient numbers are small, the as-treated analysis indicates that failed PCI procedures carried the highest risk for the patient regarding clinical endpoints; thus, achieving a successful procedure is of the highest priority.

CHALLENGE FOR RCTs OF CTO PCI. Two smaller contemporary RCTs showed similar results than this meta-analysis but with very few patients of ≤50 in each treatment arm.[44,45] Therefore, larger studies are desired to consolidate or challenge the evidence presented here. The ongoing ISCHEMIA-CTO (Revascularisation or Optimal Medical Therapy of

FIGURE 5 Survival Free of Death and Nonfatal MI With OMT or PCI



Kaplan-Meier plot of survival free of cardiovascular death and nonfatal MI in patients assigned to PCI or OMT through a 3-year follow-up according to an intention-to-treat analysis. Abbreviations as in the Figure 1.

CTO; NCT03563417) trial should add further evidence to the comparison between revascularization and OMT regarding symptomatic improvement and clinical outcomes.[46] The inclusion rules address the symptom bias observed in our study by dividing more and less symptomatic patients in different substudies. The limit for the baseline QoL is <60 points, which was observed in only 58% of our patients, and to enroll these more symptomatic patients will be a challenge. Furthermore, a follow-up may be needed beyond 3 to 5 years because a registry with a follow-up of 10 years indicated a late divergence of hard clinical endpoints.[12]

ISCHEMIA-CTO, however, will also have the limitation of being open-label, sparking the debate of whether placebo-controlled trials are required and feasible in CTO PCI. The ORBITA-2 (A Placebo-controlled Trial of Percutaneous Coronary Intervention for the Relief of Stable Angina; NCT03742050) trial had shown that such a trial was feasible at least in non-CTO lesions in CCS. In these patients, who were blinded to undergoing an interventional or placebo procedure, PCI achieved better angina control than placebo.[47] Such an approach was also planned for patients with a CTO by 2 trials, SHINE-CTO (NCT02784418), which was eventually abandoned, and ORBITA-CTO (A Placebo-controlled Trial of Chronic Total Occlusion Percutaneous Coronary Intervention for the Relief of Stable Angina; NCT05142215),[48] which has completed enrollment of 50 patients. The ethical issue is the use of deep sedation for a potentially long procedure, which is not a routine approach, and the insertion of 2 arterial sheaths with bleeding risks to simulate a CTO approach without a therapeutic intention. An alternative approach to avoid the risks of a placebo procedure but to improve the sensitivity of clinical symptom assessment could be the use of self-reporting smart devices as applied in ORBITA-2.[47]

CONCLUSIONS

In this pooled meta-analysis of the EUROCTO and DECISION-CTO trials, PCI appeared to be superior to OMT alone for symptom relief and improved QoL in patients with a single CTO and no other significant coronary lesion. This may support the offer of revascularization for improvement of symptom and QoL in severely symptomatic CCS patients with a CTO rather than long-term antianginal medication. During follow-up, there was no indication of excess harm regarding clinical events of cardiac death or MI in the PCI arm. However, we observed a wide individual variability in the symptomatic response, and not all patients randomized to PCI would benefit. The most important predictor of improvement with PCI was the severity of the baseline symptoms.

These findings based on a post-hoc analysis of a patient subset of 2 RCTs are hypothesis-generating because there were no a priori protocol harmonizations or a prespecified statistical analysis plan. Therefore, further randomized trials are needed to enhance our understanding of who would profit most from CTO PCI for symptom relief, whereas a prognostic impact in this typically low-risk population within an RCT will be difficult to prove.

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KEY WORDS chronic coronary occlusion, optimal medical therapy, percutaneous transluminal intervention, prognosis

APPENDIX For supplemental tables and figure, please see the online version of this paper.